

## **Testing of Hypothesis**

### **Hypothesis**

- A hypothesis is an assumption about relations between variables.
- Hypothesis can be defined as a logically conjectured relationship between two or more variables expressed in the form of a testable statement.
- Research Hypothesis is a predictive statement that relates an independent variable to a dependent variable.
- Hypothesis must contain at least one independent variable and one dependent variable.
- Hypothesis is an assumption about the population of the study.
- It delimits the area of research and keeps the researcher on the right track.
- Hypothesis is an assumption, that can be tested and can be proved to be right or wrong.

### **Hypothesis Testing**

- One of the important areas of statistical analysis is testing of hypothesis.
- Often, in real life situations we require to take decisions about the population on the basis of sample information.
- Hypothesis testing is also referred to as “Statistical Decision Making”.
- For Example: We may like to decide on the basis of sample data whether a new vaccine is effective in curing cold,
- whether a new training methodology is better than the existing one,
- whether the new fertilizer is more productive than the earlier one and so on.

### **Statistical Hypothesis**

Statistical hypothesis is some assumption or statement, which may or may not be true, about a population.

There are two types of statistical hypothesis

(i) Null hypothesis (ii) Alternative hypothesis

## Null Hypothesis

According to Prof. R.A. Fisher, “Null hypothesis is the hypothesis which is tested for possible rejection under the assumption that it is true”, and it is denoted by  $H_0$ .

For example: If we want to find the population mean has a specified value  $\mu_0$ , then the null hypothesis  $H_0$  is set as follows  $H_0 : \mu = \mu_0$

## Alternative Hypothesis

Any hypothesis which is complementary to the null hypothesis is called as the alternative hypothesis and is usually denoted by  $H_1$ .

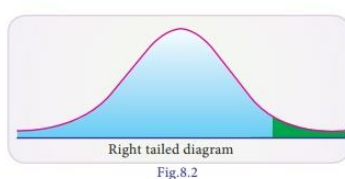
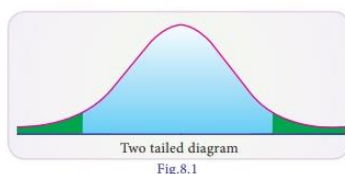
For example: If we want to test the null hypothesis that the population has specified mean  $\mu$  i.e.,  $H_0 : \mu = \mu_0$  then the alternative hypothesis could be any one among the following:

- i.  $H_1: \mu \neq \mu_0$  ( $\mu >$  or  $\mu < \mu_0$ )
- ii.  $H_1: \mu > \mu_0$
- iii.  $H_1: \mu < \mu_0$

The alternative hypothesis in  $H_1: \mu \neq \mu_0$  is known as two tailed alternative test.

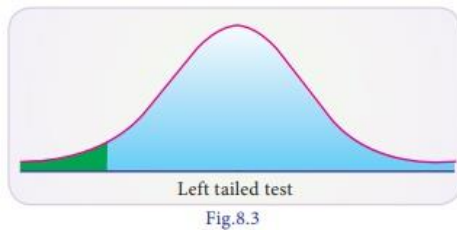
Two tailed test is one where the hypothesis about the population parameter is rejected for the value of sample statistic falling into either tails of the sampling distribution.

When the hypothesis about the population parameter is rejected only for the value of sample statistic falling into one of the tails of the sampling distribution, then it is known as one-tailed test. Here  $H_1: \mu > \mu_0$  and  $H_1: \mu < \mu_0$  are known as one tailed alternative.



Right tailed test:  $H_1 : \mu > \mu_0$  is said to be right tailed test where the rejection region or critical region lies entirely on the right tail of the normal curve.

Left tailed test:  $H_1 : \mu < \mu_0$  is said to be left tailed test where the critical region lies entirely on the left tail of the normal curve. (diagram)



### **Types of Errors in Hypothesis testing**

There is every chance that a decision regarding a null hypothesis may be correct or may not be correct. There are two types of errors. They are

Type I error: The error of rejecting  $H_0$  when it is true.

Type II error: The error of accepting when  $H_0$  it is false.

### **Critical region or Rejection region**

A region corresponding to a test statistic in the sample space which tends to rejection of  $H_0$  is called critical region or region of rejection.

### **Level of significance**

- The probability of type I error is known as level of significance and it is denoted by  $\alpha$ .
- The level of significance is usually employed in testing of hypothesis are 5% and 1%.
- The level of significance is always fixed in advance before collecting the sample information.

### **Critical values or significant values**

The value of test statistic which separates the critical (or rejection) region and the acceptance region is called the critical value or significant value. It depends upon

- (i) The level of significance

(ii) The alternative hypothesis whether it is two-tailed or single tailed.

For large samples, the standardized variable corresponding to the statistic viz.,

$$Z = \frac{t - E(t)}{\sqrt{Var(t)}} = \frac{t - E(t)}{S.E.(t)} \sim N(0,1) \text{ as } n \rightarrow \infty \quad \dots (1)$$

The value of  $Z$  given by (1) under the null hypothesis is known as test statistic. The critical values of  $Z$  at commonly used level of significance for both two tailed and single tailed tests are given in the normal probability table (Refer the normal probability Table).

Since for large  $n$ , almost all the distributions namely, Binomial, Poisson, etc., can be approximated very closely by a normal probability curve, we use the normal test of significance for large samples.

### **Testing Procedure: Large sample theory and test of significance for single mean**

The following are the steps involved in hypothesis testing problems

1. **Null hypothesis:** Set up the null hypothesis  $H_0$
2. **Alternative hypothesis:** Set up the alternative hypothesis. This will enable us to decide whether we have to use two tailed test or single tailed test (right or left tailed)
3. **Level of significance:** Choose the appropriate level of significant ( $\alpha$ ) depending on the reliability of the estimates and permissible risk. This is to be fixed before sample is drawn. i.e.,  $\alpha$  is fixed in advance.

4. **Test statistic:** Compute the test statistic

$$Z = \frac{t - E(t)}{\sqrt{Var(t)}} = \frac{t - E(t)}{S.E.(t)} \sim N(0,1) \text{ as } n \rightarrow \infty$$

5. **Conclusion:** We compare the computed value of  $Z$  in step 4 with the significant value or critical value or table value  $Z_\alpha$  at the given level of significance.

i. If  $|Z| < Z_\alpha$  i.e., if the calculated value of  $Z$  is less than critical value we say it is not significant. This may be due to fluctuations of sampling and sample data

do not provide us sufficient evidence against the null hypothesis which may therefore be accepted.

ii. If  $|Z| > Z_\alpha$  i.e., if the calculated value of  $Z$  is greater than critical value  $Z_\alpha$  then we say it is significant and the null hypothesis is rejected at level of significance  $\alpha$ .

### Test of significance for single mean

Let  $x_i, (i = 1, 2, 3, \dots, n)$  is a random sample of size  $n$  from a normal population with mean  $\mu$  and variance  $\sigma^2$  then the sample mean is distributed normally with mean

and variance  $\sigma^2/n$ , i.e.,  $\bar{x} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$ .

Thus for large samples, the standard normal variate corresponding to  $\bar{x}$  is :

$$Z = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}} \sim N(0, 1)$$

Under the null hypothesis that the sample has been drawn from a population with mean and variance  $\sigma^2$ , i.e., there is no significant difference between the sample mean ( $\bar{x}$ ) and the population mean ( $\mu$ ), the test statistic (for large samples) is:

$$Z = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}}$$

### Remark:

If the population standard deviation  $\sigma$  is unknown then we use its estimate provided by the sample variance given by  $\hat{\sigma}^2 = s^2 \Rightarrow \hat{\sigma} = s$

## Hypothesis Testing: Solved Example Problems

### Example

An auto company decided to introduce a new six-cylinder car whose mean petrol consumption is claimed to be lower than that of the existing auto engine. It was found that the mean petrol consumption for the 50 cars was 10 km per litre with a standard deviation of 3.5 km per litre. Test at 5% level of significance, whether the claim of the new car petrol consumption is 9.5 km per litre on the average is acceptable.

### *Solution:*

Sample size  $n = 50$  Sample mean  $\bar{x} = 10$  km Sample standard deviation  $s = 3.5$  km

Population mean  $\mu = 9.5$  km

Since population SD is unknown, we consider  $\sigma = s$

The sample is a large sample and so we apply Z-test

**Null Hypothesis:** There is no significant difference between the sample average and the company's claim, i.e.,  $H_0 : \mu = 9.5$

**Alternative Hypothesis:** There is significant difference between the sample average and the company's claim, i.e.,  $H_1 : \mu \neq 9.5$  (two tailed test)

The **level of significance**  $\alpha = 5\% = 0.05$

Applying the test statistic

$$Z = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}} \sim N(0,1); \quad Z = \frac{10 - 9.5}{\frac{3.5}{\sqrt{50}}} \sim N(0,1) = \frac{0.5}{0.495} = 1.01$$

Thus the calculated value 1.01 and the significant value or table value  $Z_{\alpha/2} = 1.96$

Comparing the calculated and table value, Here  $Z < Z_{\alpha/2}$  i.e.,  $1.01 < 1.96$ .

**Inference:** Since the calculated value is less than table value i.e.,  $Z < Z_{\alpha/2}$  at 5% level of significance, the null hypothesis  $H_0$  is accepted. Hence, we conclude that the company's claim that the new car petrol consumption is 9.5 km per litre is acceptable.

### Example

A manufacturer of ball pens claims that a certain pen he manufactures has a mean writing life of 400 pages with a standard deviation of 20 pages. A purchasing agent selects a sample of 100 pens and puts them for test. The mean writing life for the sample was 390 pages. Should the purchasing agent reject the manufacturer's claim at 1% level?

### Solution:

Sample size  $n = 100$ , Sample mean  $\bar{x} = 390$  pages, Population mean  $\mu = 400$  pages

Population SD  $\sigma = 20$  pages

The sample is a large sample and so we apply Z -test

Null Hypothesis: There is no significant difference between the sample mean and the population mean of writing life of pen he manufactures, i.e.,  $H_0 : \mu = 400$

Alternative Hypothesis: There is significant difference between the sample mean and the population mean of writing life of pen he manufactures, i.e.,  $H_1 : \mu \neq 400$  (two tailed test)

The level of significance  $\alpha = 1\% = 0.01$

Applying the test statistic

$$Z = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}} \sim N(0,1); Z = \frac{390 - 400}{\frac{20}{\sqrt{100}}} = \frac{-10}{2} = -5, \therefore |Z| = 5$$

Thus, the calculated value  $|Z| = 5$  and the significant value or table value  $Z_{\alpha/2} = 2.58$

Comparing the calculated and table values, we found  $Z > Z_{\alpha/2}$  i.e.,  $5 > 2.58$

**Inference:** Since the calculated value is greater than table value i.e.,  $Z > Z_{\alpha/2}$  at 1% level of significance, the null hypothesis is rejected and Therefore we concluded that  $\mu \neq 400$  and the manufacturer's claim is rejected at 1% level of significance.

### Example

(i) A sample of 900 members has a mean 3.4 cm and SD 2.61 cm. Is the sample taken from a large population with mean 3.25 cm. and SD 2.62 cm?

(ii) If the population is normal and its mean is unknown, find the 95% and 98% confidence limits of true mean.

### Solution:

(i) Given:

Sample size  $n = 900$ , Sample mean  $\bar{x} = 3.4$  cm, Sample SD  $\sigma = 2.61$  cm

Population mean  $\mu = 3.25$  cm, Population SD  $\sigma = 2.61$  cm

**Null Hypothesis**  $H_0 : \mu = 3.25$  cm (the sample has been drawn from the population mean

$\mu = 3.25$  cm and SD  $\sigma = 2.61$  cm)

**Alternative Hypothesis**  $H_1 : \mu \neq 3.25$  cm (two tail) i.e., the sample has not been drawn from the population mean  $\mu = 3.25$  cm and SD  $\sigma = 2.61$  cm.

The level of significance  $\alpha = 5\% = 0.05$

**Test statistic:**

$$Z = \frac{3.4 - 3.25}{\frac{2.61}{\sqrt{900}}} = \frac{0.15}{0.087} = 1.724$$

$\therefore Z = 1.724$

Thus, the calculated and the significant value or table value  $Z_{\alpha/2} = 1.96$

Comparing the calculated and table values,  $Z < Z_{\alpha/2}$  i.e.,  $1.724 < 1.96$

**Inference:** Since the calculated value is less than table value i.e.,  $Z < Z_{\alpha/2}$  at 5% level of significance, the null hypothesis is accepted. Hence we conclude that the data doesn't provide us any evidence against the null hypothesis. Therefore, the sample has been drawn from the population mean  $\mu = 3.25$  cm and SD,  $\sigma = 2.61$  cm.



## STUDENT'S $t$ DISTRIBUTION AND ITS APPLICATIONS

### 1. Student's $t$ -distribution

If  $X \sim N(0,1)$  and  $Y \sim \chi_n^2$  are independent random variables, then

$$T = \frac{X}{\sqrt{Y/n}}$$

is said to have  $t$ -distribution with  $n$  degrees of freedom. This can be denoted by  $t_n$ .

**Note 1:** The degrees of freedom of  $t$  is the same as the degrees of freedom of the corresponding chi-square random variable.

**Note 2:** The  $t$ -distribution is used as the sampling distribution(s) of the statistics(s) defined based on random sample(s) drawn from normal population(s).

(i) If  $X_1, X_2, \dots, X_n$  is a random sample drawn from  $N(\mu, \sigma^2)$  population then

$$X = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0,1) \text{ and } Y = \frac{\sum (X_i - \bar{X})^2}{\sigma^2} \sim \chi_{n-1}^2 \text{ are independent.}$$

Hence,

$$\begin{aligned} T_1 &= \frac{X}{\sqrt{Y/n-1}} \\ &= \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \times \frac{\sigma \sqrt{(n-1)}}{\sqrt{\sum (X_i - \bar{X})^2}} \\ &= \frac{\bar{X} - \mu}{S/\sqrt{n}} \sim t_{n-1} \quad \text{where } S = \sqrt{\frac{\sum_{i=1}^n (X_i - \bar{X})^2}{(n-1)}} \end{aligned}$$

(ii) If  $(X_1, X_2, \dots, X_m)$  and  $(Y_1, Y_2, \dots, Y_n)$  are independent random samples drawn from  $N(\mu_X, \sigma^2)$  and  $N(\mu_Y, \sigma^2)$  populations respectively, then

$$\frac{(\bar{X}-\bar{Y})-(\mu_X-\mu_Y)}{\sigma\sqrt{\frac{1}{m}+\frac{1}{n}}} \sim N(0,1) \text{ and } \frac{\sum_{i=1}^m (X_i-\bar{X})^2 + \sum_{j=1}^n (Y_j-\bar{Y})^2}{\sigma^2} \sim \chi_{m+n-2}^2 \text{ are independent.}$$

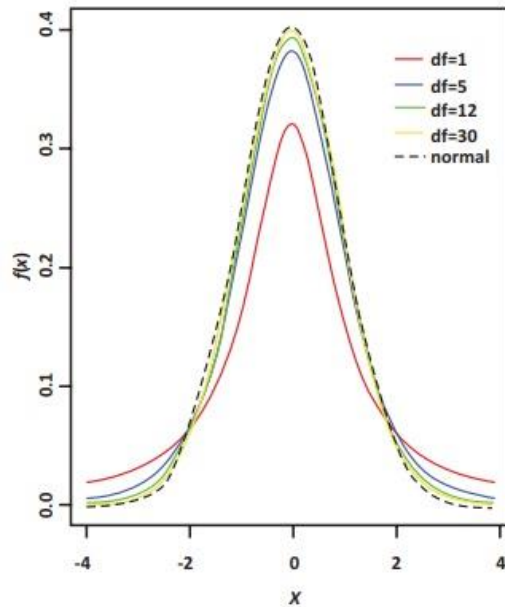
Then,

$$T_2 = \frac{(\bar{X}-\bar{Y})-(\mu_X-\mu_Y)}{S_p\sqrt{\frac{1}{m}+\frac{1}{n}}} \sim t_{m+n-2}$$

$$\text{where } S_p^2 = \frac{\sum_{i=1}^m (X_i-\bar{X})^2 + \sum_{j=1}^n (Y_j-\bar{Y})^2}{m+n-2}$$

## 2. Properties of the Student's $t$ -distribution

1.  $t$ -distribution is symmetrical distribution with mean zero.
2. The graph of  $t$ -distribution is similar to normal distribution except for the following two reasons:
  - (i) The normal distribution curve is higher in the middle than  $t$ -distribution curve.
  - (ii)  $t$ -distribution has a greater spread sideways than the normal distribution curve. It means that there is more area in the tails of  $t$ -distribution.
3. The  $t$ -distribution curve is asymptotic to  $X$ -axis, that is, it extends to infinity on either side.
4. The shape of  $t$ -distribution curve varies with the degrees of freedom. The larger is the number of degrees of freedom, closeness of its shape to standard normal distribution (fig. 1).
5. Sampling distribution of  $t$  does not depend on population parameter. It depends on degrees of freedom ( $n-1$ ).



**Figure 2.1.** Student's  $t$ -distribution

### 3. Applications of $t$ -distribution

The  $t$ -distribution has the following important applications in testing the hypotheses for small samples.

1. To test significance of a single population mean, when population variance is unknown, using  $T_1$ .
2. To test the equality of two population means when population variances are equal and unknown, using  $T_2$ .
3. To test the equality of two means – paired  $t$ -test, based on dependent samples,  $T_3$ .

### 4. Test of Hypotheses for Normal Population Mean (Population Variance is Unknown)

#### *Procedure:*

**Step 1:** Let  $\mu$  and  $\sigma^2$  be respectively the mean and variance of the population under study, where  $\sigma^2$  is unknown. If  $\mu_0$  is an admissible value of  $\mu$ , then frame the null hypothesis as

$H_0: \mu = \mu_0$  and choose the suitable alternative hypothesis from

(i)  $H_1: \mu \neq \mu_0$  (ii)  $H_1: \mu > \mu_0$  (iii)  $H_1: \mu < \mu_0$

**Step 2:** Describe the sample/data and its descriptive measures. Let  $(X_1, X_2, \dots, X_n)$  be a random sample of  $n$  observations drawn from the population, where  $n$  is small ( $n < 30$ ).

**Step 3:** Specify the level of significance,  $\alpha$ .

**Step 4:** Consider the test statistic  $T = \frac{\bar{X} - \mu_0}{S/\sqrt{n}}$ , under  $H_0$ , where  $\bar{X}$  and  $S$  are the sample mean and sample standard deviation respectively. The approximate sampling distribution of the test statistic under  $H_0$  is the  $t$ -distribution with  $(n-1)$  degrees of freedom.

$$T = \frac{\bar{x} - \mu}{s/\sqrt{n}}$$

**Step 5:** Calculate the value of  $t$  for the given sample  $(x_1, x_2, \dots, x_n)$  as

here  $\bar{x}$  is the sample mean and  $s = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2}$  is the sample standard deviation.

**Step 6:** Choose the critical value,  $t_e$ , corresponding to  $\alpha$  and  $H_1$  from the following table

| Alternative Hypothesis ( $H_1$ ) | $\mu \neq \mu_0$    | $\mu > \mu_0$     | $\mu < \mu_0$      |
|----------------------------------|---------------------|-------------------|--------------------|
| Critical Value ( $t_e$ )         | $t_{n-1, \alpha/2}$ | $t_{n-1, \alpha}$ | $-t_{n-1, \alpha}$ |

**Step 7:** Decide on  $H_0$  choosing the suitable rejection rule from the following table corresponding to  $H_1$ .

| Alternative Hypothesis ( $H_1$ ) | $\mu \neq \mu_0$               | $\mu > \mu_0$           | $\mu < \mu_0$            |
|----------------------------------|--------------------------------|-------------------------|--------------------------|
| Rejection Rule                   | $ t_0  \geq t_{n-1, \alpha/2}$ | $t_0 > t_{n-1, \alpha}$ | $t_0 < -t_{n-1, \alpha}$ |

### Example 1.

The average monthly sales, based on past experience of a particular brand of tooth paste in departmental stores is ₹ 200. An advertisement campaign was made by the company and then a sample of 26 departmental stores was taken at random and found that the average sales of the particular brand of tooth paste is ₹ 216 with a standard deviation of ₹8. Does the campaign have helped in promoting the sales of a particular brand of tooth paste?

*Solution:*

#### Step 1: Hypotheses

**Null Hypothesis**  $H_0: \mu = 200$

*i.e.*, the average monthly sales of a particular brand of tooth paste is not significantly different from ₹ 200.

**Alternative Hypothesis**  $H_1: \mu > 200$

*i.e.*, the average monthly sales of a particular brand of tooth paste are significantly different from ₹ 200. It is one-sided (right) alternative hypothesis.

#### Step 2: Data

The given sample information are:

Size of the sample ( $n$ ) = 26. Hence, it is a small sample.

Sample mean ( $\bar{x}$ ) = 216, Standard deviation of the sample = 8.

#### Step 3: Level of significance

$\alpha = 5\%$

#### Step 4: Test statistic

The test statistic under  $H_0$  is  $T = \frac{\bar{X} - \mu_0}{S/\sqrt{n}}$

Since  $n$  is small, the sampling distribution of  $T$  is the  $t$ -distribution with  $(n-1)$  degrees of freedom.

#### Step 5: Calculation of test statistic

The value of  $T$  for the given sample information is calculated from

$$t_0 = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} \text{ as}$$
$$t_0 = \frac{216 - 200}{8/\sqrt{26}} = 10.20$$

#### Step 6: Critical value

Since  $H_1$  is one-sided (right) alternative hypothesis, the critical value at  $\alpha = 0.05$  is

$$t_e = t_{n-1, \alpha} = t_{25, 0.05} = 1.708$$

#### Step 7 : Decision

Since it is right-tailed test, elements of critical region are defined by the rejection rule  $t_0 > t_e = t_{n-1, \alpha} = t_{25, 0.05} = 1.708$ . For the given sample information  $t_0 = 10.20 > t_e = 1.708$ . It indicates that given sample contains sufficient evidence to reject  $H_0$ . Hence, the campaign has helped in promoting the increase in sales of a particular brand of tooth paste.

### Example 2

A sample of 10 students from a school was selected. Their scores in a particular subject are 72, 82, 96, 85, 84, 75, 76, 93, 94 and 93. Can we support the claim that the class average score is 90?

*Solution:*

#### Step 1: Hypotheses

**Null Hypothesis**  $H_0: \mu = 90$

*i.e.*, the class average scores is not significantly different from 90.

**Alternative Hypothesis**  $H_1: \mu \neq 90$

i.e., the class means scores is significantly different from 90.

It is a two-sided alternative hypothesis.

### Step 2: Data

The given sample information is

Size of the sample ( $n$ ) = 10. Hence, it is a small sample.

### Step 3: Level of significance

$$\alpha = 5\%$$

### Step 4: Test statistic

$$\frac{\bar{X} - \mu_0}{S/\sqrt{n}}$$

The test statistic under  $H_0$  is  $T =$

Since  $n$  is small, the sampling distribution of  $T$  is the  $t$  - distribution with  $(n-1)$  degrees of freedom.

### Step 5: Calculation of test statistic

$$\frac{\bar{x} - \mu_0}{s/\sqrt{n}}$$

The value of  $T$  for the given sample information is calculated from  $t_0 =$  as under:

| $x_i$ | $u_i = x_i - A; (A = 85)$ | $u_i^2$                       |
|-------|---------------------------|-------------------------------|
| 72    | -13                       | 169                           |
| 82    | -3                        | 9                             |
| 96    | 11                        | 121                           |
| 85    | 0                         | 0                             |
| 84    | -1                        | 1                             |
| 75    | -10                       | 100                           |
| 76    | -9                        | 81                            |
| 93    | 8                         | 64                            |
| 94    | 9                         | 81                            |
| 93    | 8                         | 64                            |
|       | $\sum_{i=1}^{10} u_i = 0$ | $\sum_{i=1}^{10} u_i^2 = 690$ |

Sample mean

$$\bar{x} = A + \frac{\sum_{i=1}^{10} u_i}{n} \text{ where } A \text{ is assumed mean}$$

$$= 85 + 0 = 85$$

Sample standard deviation

$$s = \sqrt{\frac{1}{n-1} \sum_{i=1}^{10} u_i^2}$$

$$= \sqrt{\frac{1}{9} \times 690}$$

$$= \sqrt{76.67}$$

$$= 8.756$$

Hence,

$$t_0 = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$$

$$= \frac{85 - 90}{8.756/\sqrt{10}} = \frac{-5}{2.77}$$

$$= -1.806 \text{ and}$$

$$|t_0| = 1.806$$

### Step 6: Critical value

Since  $H_1$  is two-sided alternative hypothesis, the critical value at  $\alpha = 0.05$  is  $t_e = t_{n-1, \alpha/2} = t_{9, 0.025} = 2.262$

### Step 7: Decision

Since it is two-tailed test, elements of critical region are defined by the rejection rule  $|t_0| > t_e = t_{n-1, \alpha/2} = t = 2.262$ . For the given sample information  $|t_0| = 1.806 < t_e = 2.262$ .

It indicates that given sample does not provide sufficient evidence to reject  $H_0$ . Hence, we conclude that the class average score is 90.



## ***F* -DISTRIBUTION AND ITS APPLICATIONS**

*F*-statistic is the ratio of two sums of the squares of deviations of observations from respective means. The sampling distribution of the statistic is *F*-distribution.

### **Definition: *F* -Distribution**

Let  $X$  and  $Y$  be two independent  $\chi^2$  random variates with  $m$  and  $n$  degrees of freedom respectively.

$$F = \frac{X/m}{Y/n}$$

Then  $F = \frac{X/m}{Y/n}$  is said to follow *F*-distribution with  $(m, n)$  degrees of freedom. This *F*-distribution is named after the famous statistician R.A. Fisher (1890 to 1962).

### **Definition: *F*-Statistic**

Let  $(X_1, X_2, \dots, X_m)$  and  $(Y_1, Y_2, \dots, Y_n)$  be two independent random samples drawn from  $N(\mu_X, \sigma_X^2)$  and  $N(\mu_Y, \sigma_Y^2)$  populations respectively.

Then,

$$\frac{1}{\sigma_X^2} \sum_{i=1}^m (X_i - \bar{X})^2 \sim \chi_{m-1}^2 \text{ and } \frac{1}{\sigma_Y^2} \sum_{j=1}^n (Y_j - \bar{Y})^2 \sim \chi_{n-1}^2$$

are independent

(1) Hence, *F*-Statistic is defined as

$$F = \frac{(m-1)S_X^2}{\sigma_X^2} \bigg/ \frac{(n-1)S_Y^2}{\sigma_Y^2} \sim F_{m-1, n-1}$$

where

$$S_X^2 = \frac{1}{m-1} \sum_{i=1}^m (X_i - \bar{X})^2 \quad \text{and} \quad S_Y^2 = \frac{1}{n-1} \sum_{j=1}^n (Y_j - \bar{Y})^2$$

(2) F-Statistic is also defined as the ratio of two mean square errors.

### **Applications of $F$ -distribution**

The following are some of the important applications where the sampling distribution of the respective statistic under  $H_0$  is  $F$ -distribution.

- (i) Testing the equality of variances of two normal populations. [Using (1)]
- (ii) Testing the equality of means of  $k$  ( $>2$ ) normal populations. [Using (2)]
- (iii) Carrying out analysis of variance for two-way classified data. [Using (2)]

### **Test of Significance for Two Normal Population Variances**

#### ***Test procedure:***

This test compares the variances of two independent normal populations, viz.,  $N(\mu_X, \sigma_X^2)$

and  $N(\mu_Y, \sigma_Y^2)$ .

**Step 1 : Null Hypothesis**  $H_0 : \sigma_X^2 = \sigma_Y^2$

That is, there is no significant difference between the variances of the two normal populations.

The alternative hypothesis can be chosen suitably from any one of the following

(i)  $H_1 : \sigma_X^2 < \sigma_Y^2$  (ii)  $H_1 : \sigma_X^2 > \sigma_Y^2$  (iii)  $H_1 : \sigma_X^2 \neq \sigma_Y^2$

## Step 2 : Data

Let  $X_1, X_2, \dots, X_m$  and  $Y_1, Y_2, \dots, Y_n$  be two independent samples drawn from two normal populations respectively.

## Step 3 : Level of significance

$\alpha$

## Step 4 : The test Statistic

$$F = \frac{S_1^2 / \sigma_1^2}{S_2^2 / \sigma_2^2} = \frac{S_1^2}{S_2^2}$$
 under  $H_0$  and its sampling distribution under  $H_0$  is  $F_{(m-1, n-1)}$ .

## Step 5 : Calculation of the Test Statistic

$$F_0 = \frac{s_X^2}{s_Y^2}$$

The test statistic

## Step 6 : Critical values

| $H_1$                   | $\sigma_X^2 < \sigma_Y^2$  | $\sigma_X^2 > \sigma_Y^2$ | $\sigma_X^2 \neq \sigma_Y^2$                                      |
|-------------------------|----------------------------|---------------------------|-------------------------------------------------------------------|
| Critical value(s) $f_e$ | $f_{(m-1, n-1), 1-\alpha}$ | $f_{(m-1, n-1), \alpha}$  | $f_{(m-1, n-1), 1-\alpha/2}$<br>and<br>$f_{(m-1, n-1), \alpha/2}$ |

## Step 7 : Decision

| $H_1$          | $\sigma_X^2 < \sigma_Y^2$           | $\sigma_X^2 > \sigma_Y^2$         | $\sigma_X^2 \neq \sigma_Y^2$                                                       |
|----------------|-------------------------------------|-----------------------------------|------------------------------------------------------------------------------------|
| Rejection Rule | $F_0 \leq f_{(m-1, n-1), 1-\alpha}$ | $F_0 \geq f_{(m-1, n-1), \alpha}$ | $F_0 \leq f_{(m-1, n-1), 1-\alpha/2}$<br>or<br>$F_0 \geq f_{(m-1, n-1), \alpha/2}$ |

**Note 1:** Since  $f_{(m-1, n-1), 1-\alpha}$  is not available in the given  $F$ -table, it is computed as the reciprocal of  $f_{(n-1, m-1), \alpha}$ .

$$\text{i.e., } f_{(m-1, n-1), 1-\alpha} = \frac{1}{f_{(n-1, m-1), \alpha}}$$

**Note 2:** A  $F$ -test is based on the ratio of variances, it is also known as Variance Ratio Test.

**Note 3:** When  $\mu_X$  and  $\mu_Y$  are known, for testing the equality of variances of two normal populations, the test statistic is

$$F = \frac{\frac{1}{m} \sum_{i=1}^m (x_i - \mu_X)^2}{\frac{1}{n} \sum_{i=1}^n (y_i - \mu_Y)^2}$$

and follows  $F_{m, n}$ -distribution under  $H_0$

### Example 1

Two samples of sizes 9 and 8 give the sum of squares of deviations from their respective means as 160 inches square and 91 inches square respectively. Test the hypothesis that the variances of the two populations from which the samples are drawn are equal at 10% level of significance.

#### Solution:

**Step 1 : Null Hypothesis:**  $H_0 : \sigma_X^2 = \sigma_Y^2$

That is there is no significant difference between the two population variances.

**Alternative Hypothesis:**  $H_1 : \sigma_X^2 \neq \sigma_Y^2$

That is there is significant difference between the two population variances.

## Step 2 : Data

$$m = 9, n = 8$$

$$\sum_{i=1}^9 (x_i - \bar{x})^2 = 160 \quad \sum_{j=1}^8 (y_j - \bar{y})^2 = 91$$

## Step 3 : Level of significance

$$\alpha = 10\%$$

## Step 4 : Test Statistic

$$F = \frac{S_1^2 / \sigma_1^2}{S_2^2 / \sigma_2^2} = \frac{S_1^2}{S_2^2}, \text{ under } H_0.$$

## Step 5 : Calculation

$$s_X^2 = \frac{1}{m-1} \sum_{i=1}^m (x_i - \bar{x})^2 \quad \text{and} \quad s_Y^2 = \frac{1}{n-1} \sum_{j=1}^n (y_j - \bar{y})^2$$

$$s_X^2 = \frac{160}{8} = 20 \quad s_Y^2 = \frac{91}{7} = 13$$

$$F_0 = \frac{s_X^2}{s_Y^2} = \frac{20}{13} = 1.54$$

## Step 6 : Critical values

Since  $H_1$  is a two-sided alternative hypothesis the corresponding critical values are:

$$f_{(8,7),0.05} = 3.73 \quad \text{and} \quad f_{(8,7),0.95} = \frac{1}{f_{(7,8),0.05}} = \frac{1}{3.5} = 0.286$$

## Step 7 : Decision

Since  $f_{(8,7),0.95} = 0.286 < F_0 = 1.54 < f_{(8,7),0.05} = 3.73$ , the null hypothesis is not rejected and we conclude that there is no significant difference between the two population variances.

**Note 4:** The critical values of  $F$  corresponding to  $\alpha = 0.05$  requires table values at 0.025 and 0.975 which are not provided. Hence  $\alpha$  is taken as 0.1 in this example.

## Example 2

A medical researcher claims that the variance of the heart rates (in beats per minute) of smokers is greater than the variance of heart rates of people who do not smoke. Samples from two groups are selected and the data is given below. Using  $\alpha = 0.05$ , test whether there is enough evidence to support the claim.

| Smokers      | Non Smokers  |
|--------------|--------------|
| $m = 25$     | $n = 18$     |
| $s_1^2 = 36$ | $s_2^2 = 10$ |

### Solution:

**Step 1 : Null Hypothesis:**  $H_0 : \sigma_1^2 = \sigma_2^2$

That is there is no significant difference between the two population variances.

$$H_1 : \sigma_1^2 > \sigma_2^2$$

That is, the variance of heart rates of smokers is greater than that of non-smokers.

**Step 2 : Data**

| Smokers      | Non Smokers  |
|--------------|--------------|
| $m = 25$     | $n = 18$     |
| $s_1^2 = 36$ | $s_2^2 = 10$ |

**Step 3 : Level of significance**  $\alpha = 5\%$

**Step 4 : Test statistic**

$$F = \frac{S_1^2 / \sigma_1^2}{S_2^2 / \sigma_2^2} = \frac{S_1^2}{S_2^2}$$

**Step 5 : Calculation**

$$F_0 = \frac{s_1^2}{s_2^2} = \frac{36}{10} = 3.6$$

**Step 6 : Critical value**

$$f_{(m-1, n-1), 0.05} = f_{(24, 17), 0.05} = 2.19$$

**Step 7 : Decision**

Since  $F_0 = 3.6 > f_{(24, 17), 0.05} = 2.19$ , the null hypothesis is rejected and we conclude that the variance of heart beats for smokers seems to be considerably higher compared to that of the non-smokers.

**Example 3**

The following table gives the random sample of marks scored by students in two schools, A and B.

|          |    |    |    |    |    |    |    |    |    |
|----------|----|----|----|----|----|----|----|----|----|
| School A | 63 | 72 | 80 | 60 | 85 | 83 | 70 | 72 | 81 |
| School B | 86 | 93 | 64 | 82 | 81 | 75 | 86 | 63 | 63 |

Is the variance of the marks of students in school A is less than that of those in school B?

Test at 5% level of significance.

**Solution:**

Let  $X_1, X_2, \dots, X_m$  represent sample values for school A and let  $Y_1, Y_2, \dots, Y_n$  represent sample values for school B.

**Step 1 : Null Hypothesis:**  $H_1 : \sigma_X^2 = \sigma_Y^2$

That is, there is no significant difference between the two population variances.

**Alternative Hypothesis:**  $H_1 : \sigma_X^2 < \sigma_Y^2$

That is, the variance of marks in school A is significantly less than that of school B.

**Step 2 : Data**

$X_1, X_2, \dots, X_m$  are sample from school A

$Y_1, Y_2, \dots, Y_n$  are sample from school B

**Step 3 : Test statistic**

$$F = \frac{s_X^2}{s_Y^2}$$

$$s_X^2 = \frac{1}{m-1} \sum_{i=1}^m (x_i - \bar{x})^2$$

$$s_Y^2 = \frac{1}{n-1} \sum_{j=1}^n (y_j - \bar{y})^2$$

**Step 4 : Calculations**

| $x_i$ | $x_i - \bar{x}$ | $(x_i - \bar{x})^2$ | $y_i$ | $y_i - \bar{y}$ | $(y_i - \bar{y})^2$ |
|-------|-----------------|---------------------|-------|-----------------|---------------------|
| 63    | -11             | 121                 | 86    | 9               | 81                  |
| 72    | -2              | 4                   | 93    | 16              | 256                 |
| 80    | 6               | 36                  | 64    | -13             | 169                 |
| 60    | -14             | 196                 | 82    | 5               | 25                  |
| 85    | 11              | 121                 | 81    | 4               | 16                  |
| 83    | 9               | 81                  | 75    | -2              | 4                   |
| 70    | -4              | 16                  | 86    | 9               | 81                  |
| 72    | -2              | 4                   | 63    | -14             | 196                 |
| 81    | 7               | 49                  | 63    | -14             | 196                 |
| 666   |                 | 628                 | 693   |                 | 1024                |



$$\bar{x} = \frac{\sum_{i=1}^m x_i}{m} = \frac{666}{9} = 74$$

$$\bar{y} = \frac{\sum_{i=1}^n y_i}{n} = \frac{693}{9} = 77$$

$$s_X^2 = \frac{1}{9-1} \times 628 = \frac{1}{8} \times 628 = 78.5$$

$$s_Y^2 = \frac{1}{9-1} \times 1024 = \frac{1}{8} \times 1024 = 128$$

$$F_0 = \frac{78.5}{128.8} = 0.609$$

### Step 5 : Level of significance

= 5%

### Step 6 : Critical value

$$f_{(9-1,9-1),0.95} = 1/ f_{(8,8),0.05} = 1/3.44 = 0.291$$

### Step 7 : Decision

Since  $F_0 = 0.609 > f_{(8,8),0.95} = 0.291$ , the null hypothesis is not rejected and we conclude that in school B there seems to be more variance present than in school A.